



COMSATS University, Islamabad.

Department of Computer Science

Class: BSE VII

Subject: CSC334 – Parallel and Distributed Computing

Instructor: Dr. Muhammad Imran

[CLO4]: Apply DevOps pipeline automation techniques for code deployment.

Lab Task

Students have to deploy a cloud-based Employee Management web application using AWS EC2, Amazon RDS (MySQL), and Amazon S3. Students will learn how to integrate compute, database, storage, and IAM services to host a real-world application on AWS.

1. Launch and configure an EC2 instance.
2. Create and configure an Amazon RDS MySQL database.
3. Create and use an S3 bucket.
4. Connect EC2 with RDS securely.
5. Clone the Employee portal code provided on <https://github.com/hshar94/aws-live.git>
6. Deploy a Python Flask application on EC2.
7. Configure IAM roles for service access.

Solution:

1. Create EC2 machine.
2. Create RDS database.
 - a. In Aurora and RDS dashboard, select Create within Create with full configuration section.
 - b. Engine options->MySQL
 - c. Choose a database creation method->Easy Create
 - d. Templates->Free-tier
 - e. Settings
 - i. ->DB instance identifier:employee.
 - ii. ->Credentials Settings
 1. ->Master username:admin;
 2. ->Master password:adminadmin,
 3. ->Confirm master password:adminadmin
 - f. Set up EC2 connection - optional ->Connect to an EC2 compute resource.
 - g. In the EC2 instance dropdown-> select EC2 instance.
 - h. Click on create Database button.
3. Create S3 bucket
 - a. Bucket Type->General Purpose
 - b. Bucket Name->addemployee
 - c. Click on Create Bucket at the bottom.
4. Connect to EC2 machine.
 - a. `sudo apt update`



COMSATS University, Islamabad.

Department of Computer Science

Class: BSE VII

Subject: CSC334 – Parallel and Distributed Computing

Instructor: Dr. Muhammad Imran

[CLO4]: Apply DevOps pipeline automation techniques for code deployment.

Lab Task

- b. `sudo apt install mysql-client`
 - c. `mysql -h employee.cvlrhiclnrl4.us-east-1.rds.amazonaws.com -u admin -p`
 - d. `mysql> show databases;`
 - e. `mysql> create database employee;`
 - f. `mysql> use employee;`
 - g. `mysql> create table employee(`
 - h. `-> empid varchar(20),`
 - i. `-> fname varchar(20),`
 - j. `-> lname varchar(20),`
 - k. `-> pri_skill varchar(20),`
 - l. `-> location varchar(20));`
 - m. `mysql-> show tables;`
 - n. `mysql-> exit`
5. `ubuntu> git clone https://github.com/hshar94/aws-live.git`
6. `ubuntu> cd aws-live`
7. `ubuntu/aws-live> nano config.py`
- a. `customhost = "employee.cvlrhiclnrl4.us-east-1.rds.amazonaws.com"`
 - b. `customuser = "admin"`
 - c. `custompass = "adminadmin"`
 - d. `customdb = "employee"`
 - e. `custombucket = "addemployee"`
 - f. `customregion = "us-east-1"`
8. `ubuntu/aws-live>sudo apt-get install python3`
9. `ubuntu/aws-live>sudo apt-get install python3-flask`
10. `ubuntu/aws-live>sudo apt-get install python3-pymysql`
11. `ubuntu/aws-live>sudo apt-get install python3-boto3`
12. `ubuntu/aws-live>sudo python3 EmpApp.py`
13. In IAM service, create a role (EC2S3role) for EC2 for AdministrativeAccess of S3 Bucket
14. In IAM service
- a. Access Management-> Roles->CreateRole->AWS Service
 - b. ->Service or usecase->EC2->Next
 - c. ->Check Administrative Access->Next



COMSATS University, Islamabad.

Department of Computer Science

Class: BSE VII

Subject: CSC334 – Parallel and Distributed Computing

Instructor: Dr. Muhammad Imran

[CLO4]: Apply DevOps pipeline automation techniques for code deployment.

Lab Task

- d. ->RoleName-> EC2S3role->Click on createRole button.
- 15. Go to EC2 service.
- 16. Inside Actions->Security-ModifyIAMRole
- 17. Select EC2S3role->Click on UpdateRole button.
- 18. Next copy the Public IP Address (DNS) of EC2 instance to view the deployed application interface.
- 19. Fill in the form by entering the required textual information and attaching an image.
- 20. The textual data will be stored in the RDS database, while the image file will be stored in the S3 bucket.